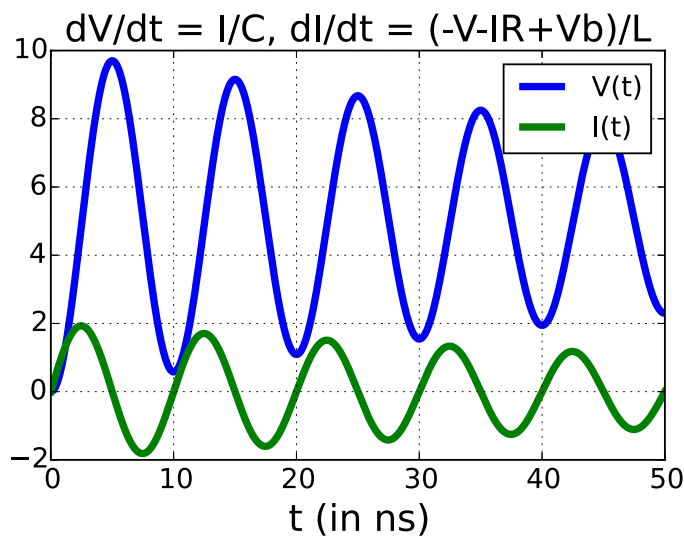
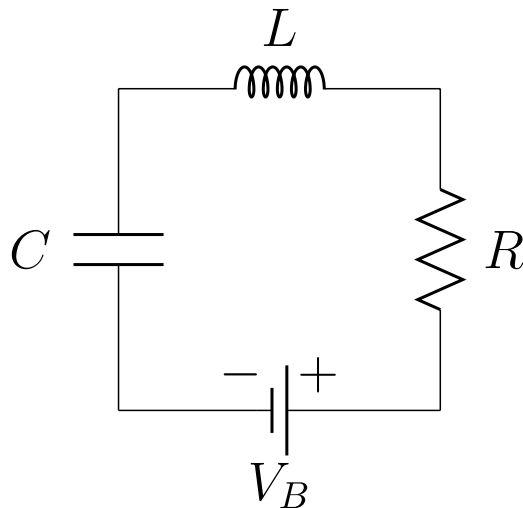


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Validation

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On the last page, you have calculated for what values this LC-oscillator will oscillate with a frequency of **100 MHz**:

$V_B = 5 \text{ V}$, $L = 0.004 \mu\text{H}$, $R = 0.1 \Omega$, and $C = 0.633 \text{ nF}$.

This choosing of the right capacitance for the condensator is called tuning. In older radios this is done mechanically with a *variable condensator*, also called a *tuning condensator*.

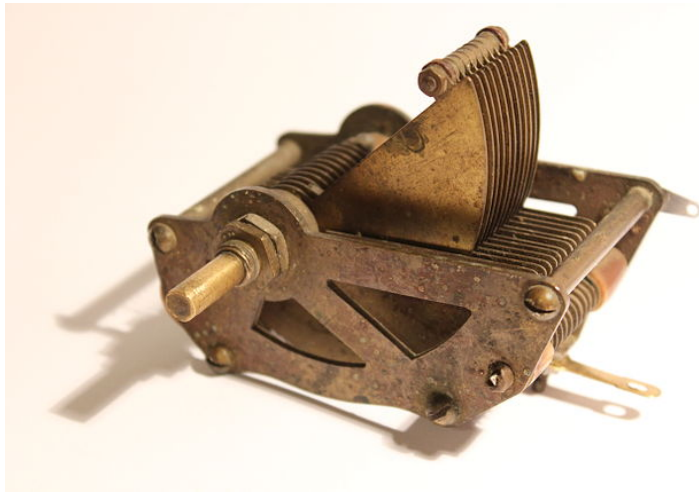


Photo: Zátonyi Sándor, (ifj.) Fized - Own work, CC BY 3.0, Link

As you can see in the graph, the amplitude of the signal dies out, because however small, there is some energy loss because $R \neq 0$ (and in reality also some charge will leak from the capacitor). So this basic LC-oscillator will only work for say **500** nanoseconds or so, which is not very long! To maintain the oscillation for longer times, other components (e.g. a transistor or an Opamp) are built into the circuit, to give the current little pushes at the right times, and keep the amplitude constant.

Oscillator circuits are not only used to set the clock speed in computers and electronic clocks and to tune radio and televisions, but also to generate those radio and television waves, and to generate the buzz of a door bell and the electronic beeps in video games.

And of course, modern oscillators are way more complicated than the one shown here. For example, often a quartz crystal is cleverly used to keep the oscillations generated by the circuit really constant over longer times. Many of the components used nowadays in circuits are non-linear, and their mathematical models contain non-linear terms and the resulting waves are often far from nice, periodic sines or cosines. Very interesting to model!

Changing the resistance R

Until now, we have kept the resistance $R = 0.1 \Omega$. In the next exercises, let's investigate what happens when R is changed.

Exercise A

1/1 point (graded)

Simulate the current in the tuned circuit for $R = 0.1 \Omega$, with $\Delta\tau = 0.001$. Just look at $I(t)$, because $V(t)$ shows similar behaviour.

Which of these words are applicable to the solution $I(t)$?

☒ bounded ✓

☐ unbounded

☐ periodic?

☒ oscillating ✓

☐ non-oscillating

☐ undamped

☒ damped ✓



Answer

Correct: Perfect, you got them all!

Submit

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

Simulate the current in the same tuned circuit but now change the resistance to $R = 10\ \Omega$.

Which of these words are applicable to the solution $I(t)$?

☒ bounded ✓

☐ unbounded

☐ periodic?

☐ oscillating

☒ non-oscillating ✓☐ undamped☒ damped ✓**Answer**

Correct: Perfect, you got them all!

Submit

You have used 2 of 3 attempts

Answers are displayed within the problem

Exercise C

1/1 point (graded)

Clearly, at some point between $R = 0.1$ and $R = 10$ the "type" of the solution changes. Find out where!

The "tipping" point lies at $R \approx$

State the answer as an integer.

5

✓ Answer: 5

5

Submit

You have used 2 of 3 attempts

Answers are displayed within the problem

Exercise D

1/1 point (graded)

Now, we go the other way, to no resistance at all. Simulate the current with Euler's method for $R = 0.0 \Omega$. You might have to increase t_{end} to for example 1000 ns to see what is happening.

Which of these words are applicable to the simulated solution $I(t)$?

☐ bounded

☒ unbounded ✓☐ periodic☒ oscillating ✓☐ non-oscillating☒ increasing amplitude ✓☐ constant amplitude☐ decreasing amplitude**Answer**

Correct: Perfect, you got them all!

You have used 2 of 3 attempts

i Answers are displayed within the problem

We saw that the value of R can change the type of solution we get. So what is going on here? All of this behaviour are effects of this system of differential equations *and* Euler's method.

In the next section you will learn more analytical techniques to investigate this kind of behaviour. In the last practice problem you can practice your new skills and find theoretical explanations for the behaviour of our simulated solutions.

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 English ▼

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